AI PYQ 2025 — ANSWER KEY

Q.P. Code 104 • Set 4 • MCQs, Short Answers & Long Answers

Note for students: This answer key follows the extracted question text available from the supplied paper. Items dependent on diagrams or unclear textbook terminology are explicitly flagged for verification.

Section 1 — MCQs

Q1. Employability Skills

Subquestion	Answer	Explanation
(i)	D — Negative thoughts	Negative thinking increases stress rather than helping manage it.
(ii)	C — Junk mails	Spam refers to unwanted or unsolicited emails.
(iii)	A — Both A and R are correct, and R explains A	Sustainable agriculture protects the environment; reducing harmful chemical use can help protect soil.
(iv)	C — Being indisciplined	Self-motivation requires discipline and consistent effort.
(v)	D — Small group communication	A small group can exchange ideas and coordinate effectively.
(vi)	B — Creative	Reema develops new ideas for dishes and her business.

Q2. Artificial Intelligence Fundamentals

Subquestion	Answer	Explanation
(i)	C — A is correct but R is incorrect	Mimicking human traits is associated with AI, but a fully automatic washing machine is not necessarily AI.
(ii)	C — Artificial Intelligence	Recommendation systems use AI to suggest content and products based on user preferences.
(iii)	D — Statement 2 is correct but Statement 1 is incorrect	Stakeholders are people affected by a problem or who benefit from its solution. In the 4Ws canvas, they are identified under Who, not When.
(iv)	C — Both (A) and (B)	An AI model is trained using data and tested to evaluate its performance.
(v)	B — Extracting meaningful information from images	Image processing helps computers analyse image content.
(vi)	A — Stop word	Common words such as “an”, “is” and “the” are often treated as stop words.

Q3. Data Science, Computer Vision and NLP

Subquestion	Answer	Explanation
(i)	Likely Google Maps	The answer was inferred from the extracted question context; verify the logo/figure on page 8.
(ii)	D — Searching for a relative’s house address on the Internet	This is not an application of genetics or genomics.
(iii)	B — Classification + Localization	This identifies an object’s class and its location in an image.
(iv)	A — SQL	Structured Query Language is designed to manage and query structured data in databases.
(v)	D — Automatic Summarization	It creates a concise overview of longer text while reducing repetition.
(vi)	Needs diagram verification	The visual on page 10 is required to identify the correct evaluation condition.

Q4. Model Evaluation, Learning and Data Formats

Subquestion	Answer	Explanation
(i)	B — Comma Separated Values	CSV stores tabular data as values separated by commas.
(ii)	A — Both statements are correct	Overfitting is undesirable because a model may memorise training data instead of generalising well.
(iii)	A — Precision	Precision is the proportion of positive predictions that are actually positive.
(iv)	Needs diagram verification	The diagram is needed to distinguish clustering, classification, regression or dimensionality reduction.
(v)	C — Text Classification	Spam filtering classifies emails into categories such as spam and non-spam.
(vi)	B — Internet of Things (IoT)	Internet-connected devices can be monitored or controlled remotely.

Q5. Musical Intelligence and AI Applications

Subquestion	Answer	Explanation
(i)	D — Recognising and creating sounds, rhythms and sound patterns	These abilities define musical intelligence.
(ii)	B — True Positive and True Negative	In both cases, the prediction agrees with reality.
(iii)	C — Statement 1 is correct but Statement 2 is incorrect	Image-based search uses visual information, not sounds.
(iv)	Verify expected stemming method	The extracted answer previously suggested “Study”, but stemming algorithms can produce “studi”; check the prescribed method/textbook.
(v)	A — Viral Disease Outbreak	A false negative can miss an infected person, allowing disease to spread.
(vi)	AI-powered chatbot using a database (exact term to verify)	The question describes a flexible chatbot that works with a large database.

Section 2 — Short Answers

Q6. Explain the importance of a healthy lifestyle in managing stress. Give one common cause of stress among children.

A healthy lifestyle, including proper sleep, nutritious food and regular exercise, helps reduce stress and improve concentration. Academic pressure is a common cause of stress among children.

Q7. Mention two methods for effective communication with a team of 20 people.

1. Team meetings: conduct regular meetings to share updates and discuss problems.
2. Digital communication: use emails or group messaging platforms to share instructions and important information.

Q8. Write any two tasks entrepreneurs perform while running their business.

Entrepreneurs plan business activities, organise resources, manage finances, develop products or services, attract customers and identify opportunities for growth.

Q9. Mention two measures to protect digital devices.

1. Install security updates and antivirus software.
2. Use strong passwords and avoid downloading files from untrusted websites.

Q10. Discuss the problems of sustainable development related to water and fuel.

Water: Water scarcity, pollution and wastage reduce safe water availability for future generations.
Fuel: Excessive fossil-fuel consumption depletes resources and increases air pollution and greenhouse gas emissions.

Q11. Differentiate between Computer Vision (CV) and Natural Language Processing (NLP).

Computer Vision: Enables computers to interpret images and videos; example: face recognition.
NLP: Enables computers to understand and process human language; example: chatbots and translation.

Q12. Define Data Exploration and Data Features in the AI Project Cycle.

Data Exploration: Examining collected data to understand its structure, patterns, relationships and errors before developing an AI model.
Data Features: Characteristics or attributes used by an AI model to learn patterns and make predictions.

Q13. Give two uses of Data Science in airline route planning.

1. Identify efficient routes that reduce flight distance, travel time and fuel consumption.
2. Forecast passenger demand to optimise schedules, aircraft allocation and route planning.

Q14. Give two key impacts of Computer Vision on medical imaging.

1. Helps detect abnormalities such as tumours in medical images.
2. Assists doctors in analysing scans more efficiently and consistently, supporting diagnosis and treatment planning.

Q15. What is the primary difference between Human Language and Computer Language?

Human language is natural, flexible and often ambiguous. Computer language follows defined syntax and rules so computers can interpret instructions and execute operations.

Q16. Explain False Positives and False Negatives when detecting fraudulent financial transactions.

False Positive (FP): A genuine transaction is incorrectly classified as fraudulent.
False Negative (FN): A fraudulent transaction is incorrectly classified as genuine and may go undetected.

Section 3 — Long Answers

Q17. Explain AI Bias and AI Access. Give one example of each. (4 marks)

AI Bias occurs when an AI system produces unfair or systematically inaccurate results for certain individuals or groups. It can arise from unrepresentative training data, historical inequalities or biased design decisions.

Example: A recruitment AI trained mainly on applications from one demographic group may unfairly disadvantage equally qualified applicants from other groups.

AI Access refers to people's ability to access and benefit from AI technologies. Access can be limited by cost, internet connectivity, digital literacy, language barriers or lack of suitable devices.

Example: Students in remote villages with poor internet connectivity may not have the same opportunities to use AI-powered learning platforms as students in well-connected cities.

Therefore, AI should be designed to minimise bias and make its benefits accessible to diverse groups of people.

Q18. Explain the use of the problem statement template with respect to the 4Ws. Draw a template. (4 marks)

The problem statement template converts information collected through the 4Ws of problem scoping into a clear, focused statement. It helps identify the people affected, the problem they face, where and why it occurs, and the benefit expected from a solution.

4W element	Key question
Who	Who experiences the problem?
What	What is the problem?

4W element	Key question
Where	Where does the problem occur?
Why	Why is it important to solve?

Combined template: Our target users (Who) face [specific problem] (What) in [location or situation] (Where), and solving it is important because [reason/benefit] (Why).

Example: Students in a school struggle to separate recyclable waste from general waste because clearly labelled disposal facilities are unavailable. A waste-segregation system could improve cleanliness and recycling.

Q19. Identify the concept shown in the diagram and explain its working. (4 marks)

The diagram required for this question was not visible in the extracted text of the supplied PDF. The following is a conditional example only; verify the original diagram before using it.

If the diagram represents an Artificial Neural Network (ANN): An ANN is a machine-learning model inspired by interconnected neurons in the human brain. It consists of an input layer, one or more hidden layers and an output layer.

1. The input layer receives data, such as numerical values or image features.
2. Hidden layers process inputs using weights, biases and activation functions.
3. During training, the network compares predictions with actual outputs and adjusts its weights to reduce errors.
4. The output layer produces the final prediction or classification.

ANNs are used in image recognition, speech processing and prediction tasks.

Q20. Implement all four steps of Bag of Words (BoW) to create a document vector table. (4 marks)

Document 1: NLP is a domain of AI.
Document 2: NLP stands for Natural Language Processing.

Step 1 — Collect documents

Write down both documents as provided.

Step 2 — Tokenisation

Split each document into words and remove punctuation.

Document 1: NLP, is, a, domain, of, AI
Document 2: NLP, stands, for, Natural, Language, Processing

Step 3 — Create vocabulary

Combine unique words from both documents. For consistency, convert words to lowercase.

Vocabulary: nlp, is, a, domain, of, ai, stands, for, natural, language, processing.

Step 4 — Create document vector table

Word	Document 1	Document 2
nlp	1	1
is	1	0
a	1	0
domain	1	0
of	1	0
ai	1	0
stands	0	1
for	0	1

Word	Document 1	Document 2
natural	0	1
language	0	1
processing	0	1

A value of 1 means the word appears in the document; 0 means it does not. The resulting vectors represent the documents numerically.

Q21. Calculate True Negative, Precision, Recall and F1 Score. (4 marks)

Confusion matrix from the question:

Prediction	Reality: Yes	Reality: No
Yes	110 (TP)	60 (FP)
No	50 (FN)	410 (TN)

(A) True Negative: A true negative is a case predicted as negative when reality is also negative. Therefore, **TN = 410 cases**.

(B) Precision

Precision = TP / (TP + FP) = 110 / (110 + 60) = 110 / 170 = **64.71%**.

Recall

Recall = TP / (TP + FN) = 110 / (110 + 50) = 110 / 160 = **68.75%**.

F1 Score

F1 = 2TP / (2TP + FP + FN) = 220 / (220 + 60 + 50) = 220 / 330 = **66.67%**.

Parameter	Answer
True Negative	410 cases
Precision	64.71%
Recall	68.75%
F1 Score	66.67%

Interpretation: Precision is the proportion of predicted positive cases that are actually positive. Recall is the proportion of actual positive cases detected. The F1 Score balances precision and recall.